EG1298

Isochronous Speed Engine Controller

HIGHLIGHTS

- **Isochronous speed governing with paralleling support**
- **Configurable load based droop governing**
- **MODBUS over RS485 for remote communication**
- **Manually configurable via front fascia or via USB port using SEDEMAC's configuration utility**
- **Back-lit and full graphics display with power saving feature**

EG1298 Isochronous Electronic Governor

SEDEMAC's EG1298 is an isochronous engine speed governing solution which provides **isochronous speed governing** feature with an option to switch between load based droop governing and paralleling. It also offers engine safety and load monitoring.

The EG1298 controller is a modern engine controller for genset application with user-friendly HMI and full graphics LCD. It works with SEDEMAC'S rotary actuators for isochronous governing.

MODBUS over **RS485** is provided for supporting the remote communication.

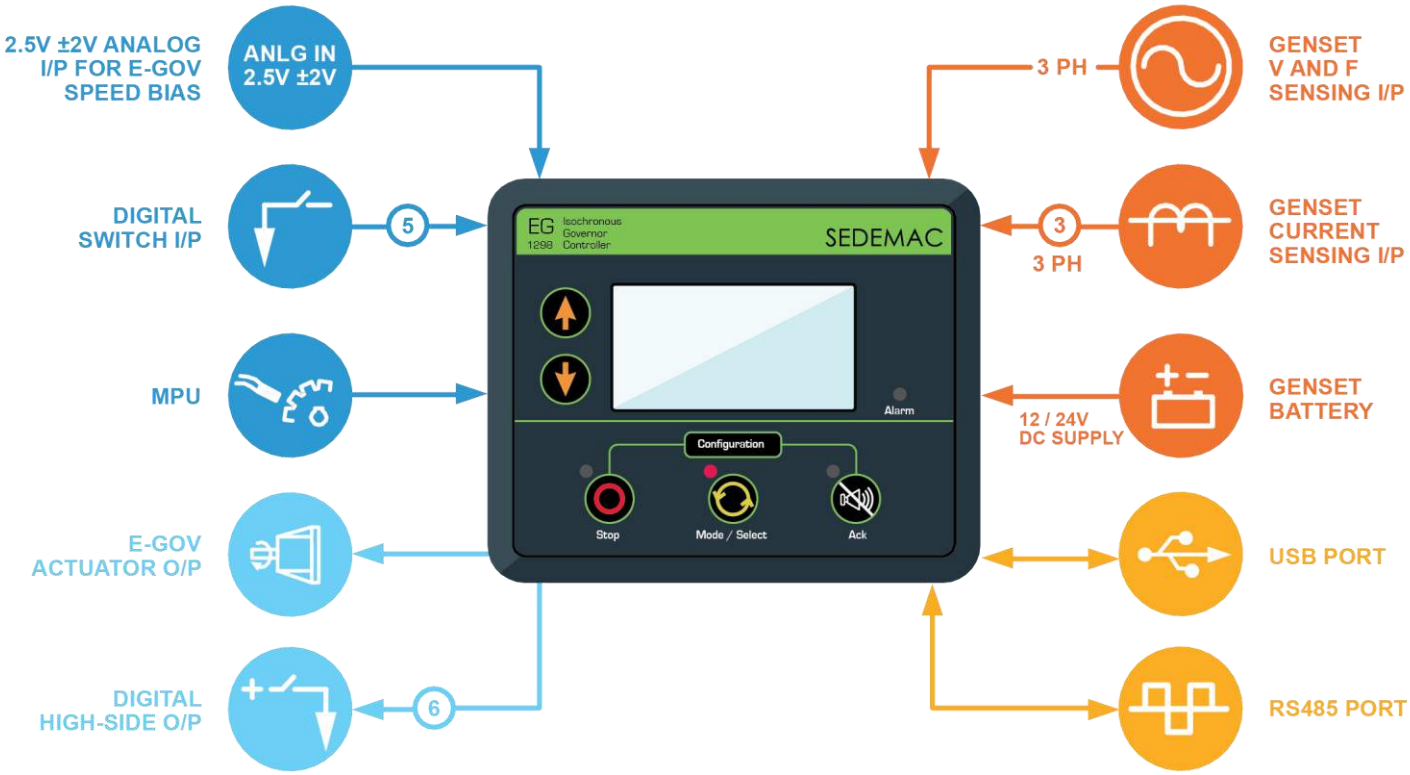
The device has an **USB** (Type-B) port that empowers it to communicate with SEDEMAC's proprietary Microsoft Windows based utility "**SEDEMAC-Smart Config**".

Real Time Clock (RTC) is provided for accurate event logging (100 counts) and data analysis.

Provision is there to Interface **Magnetic Pickup Unit** (MPU) to read the Engine RPM.

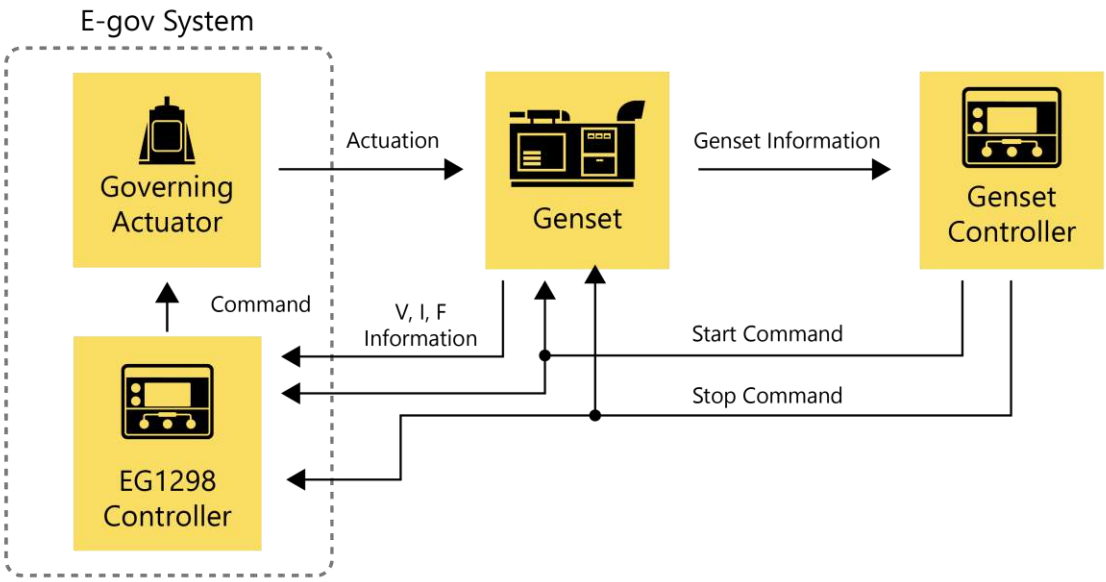
EG1298

SCHEMATIC »



Isochronous Speed Engine Controller

SYSTEM LAYOUT



FEATURES »

- Provides isochronous speed governing
- Offers load based droop governing
- Supports paralleling with 2.5 ± 2V input
- Full graphics back-lit LCD display with power saving mode
- Monitors engine safety parameters – over speed, under speed and emergency stop
- Supports magnetic pick-up unit interface for engine speed sensing
- Versatile software capable of handling "events" during operation; event log for 100 events with RTC stamps
- Totalizers for engine starts, running hours, kWh, kVAh
- MODBUS over RS485 is used for communication with other devices
- Supports end-of-line parameter flashing at customer's site
- Offers multiple level passwords to avoid tampering with configuration settings
- Access to each parameter through SEDEMAC – Smart Config or control keys of EG1298 controller

Ordering information

EG1298 controller: SM0003884

Control Panel Details:

- CT
- Panel Enclosure
- Wiring
- Relays

Kindly refer SEDEMAC's datasheet of "Rotary Actuators for Electronic Speed Governing" for interfacing actuator with the controller.

SPECIFICATIONS »

Environmental Testing Standards

Operating/Storage Temperature
-20˚C to 65˚C/-30˚C to 70˚C , as per IEC 60068-2-1, 2

Vibration
2G in X, Y and Z axes, as per IEC 60068-2-6

Humidity
0% to 95% RH, as per IEC 60068-2-78

EMI/EMC
Compliance as per IEC 61000-6-2, 4

Power Supply
8V - 32V DC continuous; power transients, as per ISO 7637 - 2

Ingress Protection
IP65 for front face with optional gasket, as per IEC 60529

Shock
15 g for 11 ms, compliance as per IEC 60068-2-27

Technical

Maximum Standby Current
~0.08A @12V or 24V DC

Digital Output
6 x 0.75A high-side drivers configurable for over speed, under speed, engine running etc.

Digital Input
5 switch-to-ground inputs configurable for engine start, engine stop, emergency stop

Analog Input
1 analog input (2.5 ± 2V) for E-Gov speed bias

E-governor Module Interface
1A, 2-phase output for Rotary Actuator
2.5 ± 2V input for target speed bias; 2.5V for zero bias

Genset Current Measurement
0A – 5A AC RMS for Current Transformer (CT) secondary

Magnetic Pickup Measurement
0.2V – 45V AC RMS @10 Hz – 10 kHz

EG1298 Controller Dimensions
External dimensions: 139 mm x 114 mm x 37.5 mm
Mounting panel cut: 118 mm x 93 mm

SEDEMAC GENSET CONTROLS »

Genset Controllers

Electronic Governors

ATS Controllers

R2D3 IIoT Platform

Battery Chargers

www.sedemac.com

EG1299

Variable Speed Engine Controller for Telecom Backup Gensets

HIGHLIGHTS »

- **Built-in electronic governor module**
- **Variable and fixed speed operation**
- **Configurable 0% droop with paralleling support**
- **MODBUS over RS485 for remote communication**
- **Manually configurable via front fascia or via USB port using SEDEMAC's configuration utility**
- **Backlit and full graphics display with power saving feature**

EG1299 Variable Speed Electronic Governor

SEDEMAC'S EG1299 is an engine controller which provides **variable speed governing** feature with an option to switch over 0% droop governing with paralleling support.

The EG1299 is a modern engine controller for telecom applications with user-friendly HMI and full graphics LCD.

The EG1299 controller offers **engine safety parameters** (over speed, under speed and emergency stop) and **load monitoring.**

MODBUS over RS485 is provided for supporting remote communication.

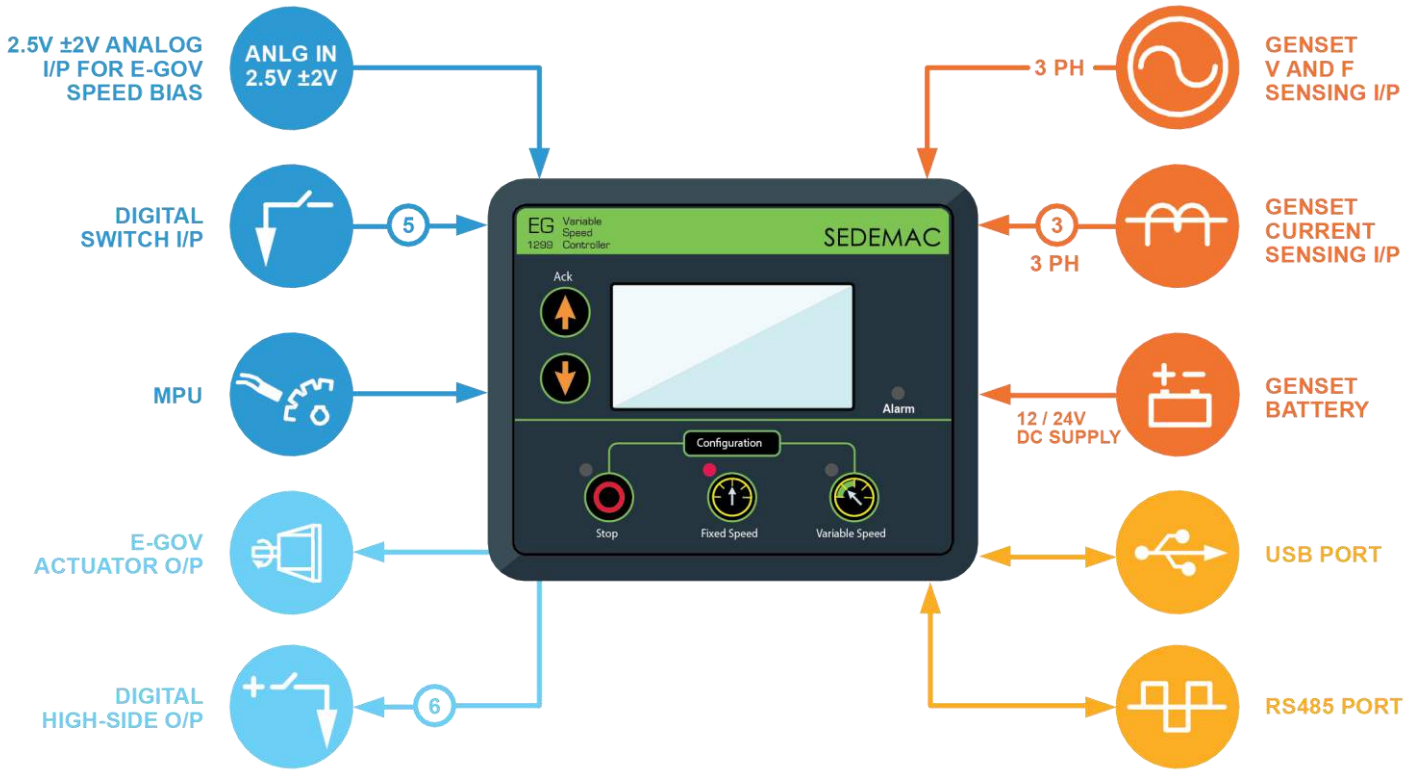
The device has an **USB** (Type-B) port that empowers it to communicate with SEDEMAC's Proprietary Microsoft Windows based utility "**SEDEMAC-Smart Config**".

Real Time Clock (RTC) is provided for accurate event logging (100 counts) and data analysis.

Provision is there to interface **Magnetic Pickup Unit** (MPU) to read the Engine RPM.

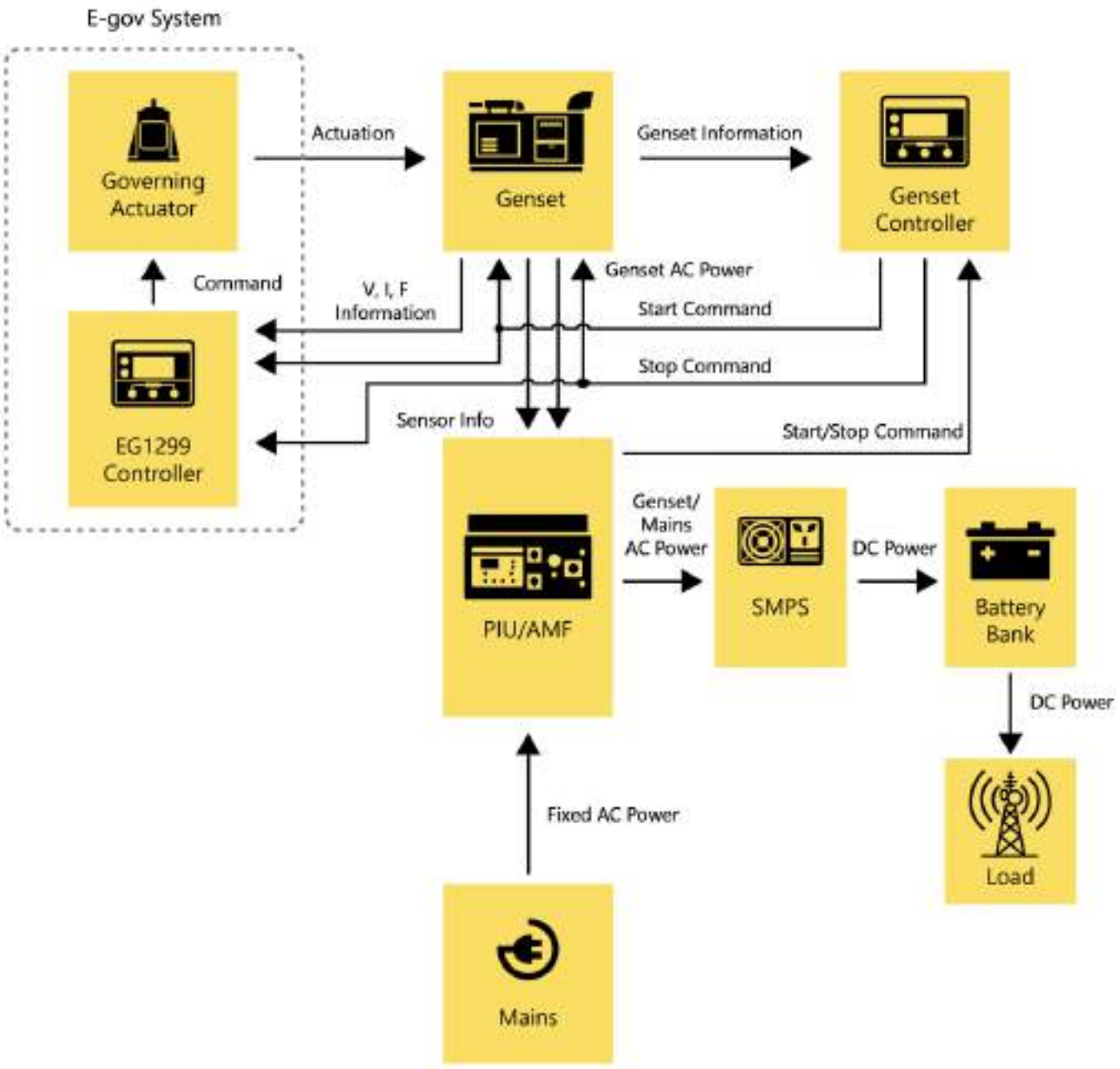
EG1299

SCHEMATIC »



Variable Speed Engine Controller

SYSTEM LAYOUT



Engine RPM and Fuel Consumption Analysis

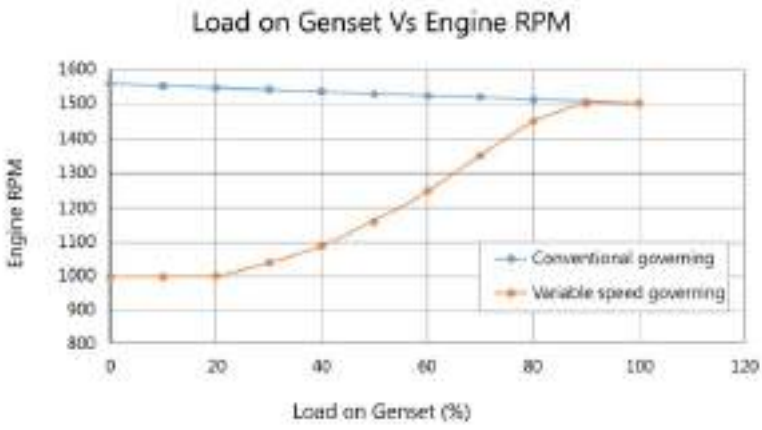


Figure 1: Engine RPM Vs Load analysis for conventional governing and variable speed governing

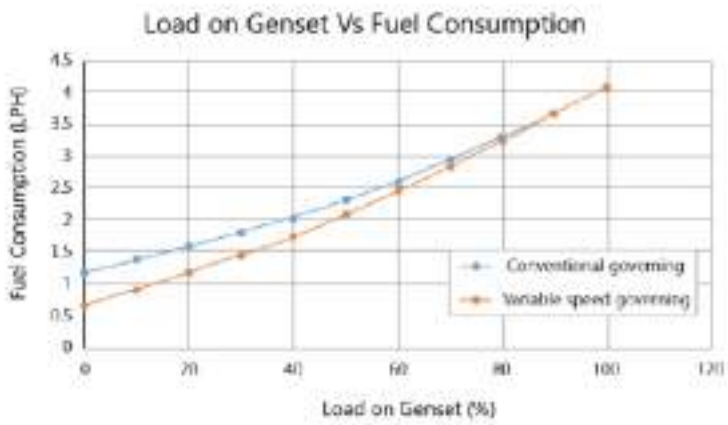


Figure 2: Fuel consumption analysis for conventional governing and variable speed governing

- In conventional governing, engine's speed drops down up to 4% of it's set value with increment in load application, as shown in figure 1.
- In variable speed governing, engine's speed is varied with respect to load. Lesser the load, lower will be the speed of the engine. This load vs speed mapping promises significant reduction in fuel consumption as shown in figure 2.
- The optimum RPM for a given load is derived through a fuel mapping exercise, where the engine is run at different RPM at the specified loads. The analysis is made for the best specific fuel consumption at the given load.

FEATURES »

- Provides variable speed governing as per load on genset
- Easy shift from variable speed governing feature to fixed speed governing through key/digital input
- Full graphics back-lit LCD display with power save mode
- Monitors engine safety parameters – over speed, under speed and emergency stop
- Supports the MPU interface for engine speed sensing
- Versatile software capable of handling "events" during operation; Event log for 100 events with real time clock (RTC) stamps
- Built-in electronic governor module; directly connects to SEDEMAC's Rotary Actuators* to guarantee ISO8528: G3 class performance for 0% droop mode; accepts target speed bias input (2.5 ± 2V) from an external load sharing module (LSM)
- Totalizers for engine starts, running hours, kWh, kVAh, etc.
- PC connectivity via USB port and RS485 via MODBUS
- Supports end-of-line parameter flashing at customer's site
- Offers multiple level passwords for avoiding tampering of configuration
- Access to each parameter through SEDEMAC – Smart Config or control keys of EG1299 controller

Ordering Information:

EG1299 controller.....SM0001958

Control Panel Details:

- CT
- Panel Enclosure
- Wiring
- Relays

**Kindly refer SEDEMAC's datasheet of "Rotary Actuators for Electronic Speed Governing" for interfacing actuator with the controller*

SPECIFICATIONS »

Environmental Testing Standards

Operating/Storage Temperature
-20 to 65˚C/-30˚C to 70˚C, as per IEC 60068-2-1, 2

Vibration
2G in X, Y and Z axis, as per IEC 60068-2-6

Humidity
0% to 95% RH, as per IEC 60068-2-78

EMI/ EMC
Compliance as per IEC 61000-6-2, 4

Power Supply
8V – 32V DC continuous; power transients, as per ISO 7637-2

Ingress Protection
IP65 for front face with optional gasket, as per IEC 60529

Shock
15 g for 11 mS, compliance as per IEC 60068-2-27

Technical

Maximum Standby Current
~0.08A @ 12 or 24V DC

Digital Output
6 x 0.75A high-side drivers configurable for over speed, under speed, engine running etc.

Digital Input
5 switch-to-ground inputs configurable for engine start, engine stop, emergency stop

Analog Input
1 analog input (2.5 ± 2V) for E-Gov speed bias

E-governor Module Interface
1A, 2-phase output for Rotary Actuator
2.5 ± 2V input for target speed bias; 2.5V for zero bias

Load Current Measurement
0 – 5A AC RMS for Current Transformer (CT) secondary

Magnetic Pickup Measurement
0.2V – 45V AC RMS @10 Hz – 10 kHz

EG1299 Controller Dimensions
External dimensions: 139 mm x 114 mm x 37.5 mm
Mounting panel cut: 118 mm x 93 mm

SEDEMAC GENSET CONTROLS »

Genset Controllers

Electronic Governors

ATS Controllers

R2D3 IIoT Platform

Battery Chargers

B12 ECU

Electronic Control Unit for Isochronous governing

HIGHLIGHTS

Offers Isochronous Governing within ISO 8528: G3 class

Supports paralleling through external Load Sharing Module

Option for providing RPM or time based Low Side driver output

B12 ECU

SEDEMAC's B12 Electronic Control Unit is designed and customized for precision motion control on wide variety of diesel/gas/gasoline engines with mechanical fuel systems.

It provides Isochronous Governing, Low Side driver output (on/off type) based on RPM, and time.

Apart from Isochronous governing, it offers premium features such as enabling Variable Speed governing, genset paralleling.

EXTERNAL INTERFACE

INPUT 1: Speed signal from inductive/magnetic crank speed sensor

INPUT 2: Engine STOP signal (energize to run/ stop) from genset panel/controller

INPUT 3: For paralleling mode: 12/24 V DC auxiliary signal + speed bias signal (2.5V ± 2V analog)

POWER: Supply from 12V/24V battery

OUTPUT 1: To actuator

OUTPUT 2*: Low Side Driver (LSD) o/p RPM or time based

**Output 2 is provided in different hardware variants*

This output is offered in SEDEMAC's ECU with isochronous governing & Low Side driver output hardware.

FEATURES & BENEFITS »

- Isochronous Governing*: Meets G3 class governing parameters as per ISO 8528
- **Genset Paralleling**: Supports paralleling/synchronization of two gensets in collaboration with an external LSM, taking a 2.5 ± 2V input speed bias
- Supports paralleling with other governors or even mechanical governors
- **Variable Speed Operation**: Supports variable speed operation based on externally generated analog signal (0-5V) as the target governing speed
- **Plug and play**: All electronic control unit come pre-programmed from SEDEMAC based on target engine platform. No further calibration/PID gains setting needed at the engine-OEM plant. The calibration needs to be done once for productization.

Ordering Information:

ECU..... SM0000268

*For Isochronous and Variable Speed Governing, use SEDEMAC's Rotary Actuator for controlling the fuel control lever. Kindly refer SEDEMAC's datasheet of "Rotary Actuators for Electronic Speed Governing" for specifications of the Actuator variants.

SPECIFICATIONS »

Environmental

Temperature
-20°C to 85°C as per IEC 60068-2-1, 2

Vibration
5G in X, Y and Z axes for 10 Hz - 500 Hz, as per IEC 60068-2-6

Humidity
0% to 95% RH as per IEC 60068-2-78

EMI/EMC
Compliance as per ISO 11452-2, CISPR 25-Level1

Power Supply Aberration
Reverse and double polarity protection as per JASO D094-01

Ingress Protection
IP 65 as per IEC 60068-2-27

Technical

Power Consumption
Less than 1A @ 12V/24 V DC
LSD output with 2A current rating

Construction
Nylon-6 with 30% Glass Filled casing

Electrical Interface
12-pin automotive sealed connector

Software Features
Closed loop engine speed control with special modes for speedy cold start, engine stop, recovery from block Loading. Support for variable speed governing, genset paralleling

Weight: 180 g

Dimensions L x W x H : 96mmx 90mm x 30mm

Performance

Governing Performance: ISO 8528: G3 class. Zero droop
Low Side Driver Output: RPM or time based

Variable Speed Governing: Controls the engine speed as per Load Vs RPM lookup table, taking a 0 - 5V analog input as the load reference

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Rotary Actuators

for Electronic Speed Governing

HIGHLIGHTS »

Precision motion combined with high torque density and low power consumption

Control capability for all types of fuel injection pumps

Robust construction for engine mounted application

Rotary Actuator

SEDEMAC's Rotary Actuators are designed and customized for precision **motion control application** on wide variety of diesel/gas/gasoline engines with mechanical fuel injection systems.

Its **robust architecture** offers high rigidity and strength to sustain against harsh environmental conditions and mechanical vibrations for engine mounted applications.

Powered by SEDEMAC's B12 Series governor ECUs or with SEDEMAC's genset controllers for the genset market, the Actuator helps delivering tight **speed governing performance** within **ISO 8528:G3 class** limits.

FEATURES »

External Interface

- **Power Supply**
  Through SEDEMAC's B12 ECU / Genset controllers*
- **Input**
  4-wire power interface
- **Output**
  Rotary motion of the o/p shaft

* The Rotary Actuators are to be operated only by SEDEMAC's B12 Series governor ECUs or Genset controllers.

Feature List

Standard Features

- 150 deg travel with motion limit studs at either ends
- D-Cut on the actuator o/p shaft for angular alignment of the linkage

Optional Features

- Return spring integration (optional) for default zero-fuel low idle position**

Ordering Information:

Standard Torque SM0000015
Standard Torque with return Spring SM0000014
Medium Torque..... SM0003648
Medium Torque with return spring..... SM0000020
High Torque..... SM0001956
High Torque with return spring..... SM0000021

** The return spring is unidirectional, stopping direction is anticlockwise when seen from Actuator's shaft face

SPECIFICATIONS »

Environmental

Operating/Storage Temperature
-20˚C to 85˚C as per IEC 60068-2-1, 2

Vibration
5G in X, Y and Z axes for 8-500 Hz as per IEC 60068-2-6

Humidity
0% to 95% RH as per IEC 60068-2-78

Ingress Protection
IP56 as per IEC 60529

Technical

Power Consumption
1A

Motion
150 deg travel

Electrical Interface
4-wire connector

Construction
Plastic casing (Nylon-6 with 30% glass filled)

Types Based on Output Torque
Standard Torque: 0.35 Nm
Medium Torque: 0.56 Nm
High Torque: 0.93 Nm

Weight
Standard Torque: 870 g
Medium Torque: 1270 g
High Torque: 1600 g

Dimensions
Standard Torque: 94 mm x 84 mm x 85 mm
Medium Torque: 94 mm x 84 mm x 121 mm
High Torque: 94 mm x 84 mm x 140 mm

Mounting Foot Print Dimension (For All Variants)
78 mm x 40 mm
4 Number of holes - 6.7 mm Diameter each

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About Company

SEDEMAC Mechatronics is a mechatronic products company engaged in the business of control solutions for automotive and off-highway powertrains. The company manufactures a wide range of control products like a variety of AMF genset controllers, electronic governors and variable speed controls for diesel gensets / engines. SEDEMAC's portfolio for gasoline engines / powertrains includes products like Smart Ignition, Integrated Starter Generator (ISG), ECU, Automated Manual Transmissions (AMT) ECU, Electronic Fuel Injection (EFI) ECU and so on.

A group of researchers founded SEDEMAC in 2007 in Mumbai, India. What started as a small lab-based enterprise has now grown into a rapidly growing company that supplies / licenses unique control products to the leading OEMs in India.

Each of the SEDEMAC's products goes through complete cycle of need of identification, development of relevant technology, design of practical products around developed technology, validation of products coordinating with key potential customers, organization of supply chain for series production, delivery to customers and effective after sales product support.

SEDEMAC is #1 in genset market control in India today and is the preferred choice of leading OEM's for unique control products.

Our Facilities

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